SCHEME and ANSWER KEY for QP Code: 25901440

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

Fourth Semester UG (Honours) Examination – March 2026

Course: MG4DSEPHY201 – Numerical Methods for Computational Physics

(2024 Admission onwards)

Duration: 1 hour 30 minutes **Maximum Marks:** 50

PART A – Short Answer Questions

Answer any 10 questions. Each question carries 2 marks. [10 × 2 = 20 marks]

Q.No.	Key Points	Marks
1	After converting the system of equations to upper triangular form, variables are solved starting from the last equation upward. OR explanation with example	1 1 2
2	Simple root ⇒ occurs once Multiple root ⇒ root repeated OR explanation with example	1 1 2
3	The interval between consecutive independent variable values. OR ($h = x_{i+1} - x_i$). Interpolation- definition/ idea	2 1
4	Line ($y = 2x + 3$). A straight increasing line with slope 2 and intercept 3. Approximate graph	2
5	NR: fast (quadratic convergence), needs derivative, may diverge. Division by zero may occur Bisection: slower, derivative-free, always convergent. (Any two points).	2
6	$\nabla B = B_{15} - B_{10} = 105.1 - 102.8 = 2.3$ Definition/equation Substitution Final result	2 1 0.5 0.5
7	Used when x-values are **equally spaced** and the **value to be found is near the beginning** of the table. (any one point, but the 2nd point is the most suitable)	2
8	To organize function values and finite differences systematically for interpolation/extrapolation (and numerical differentiation). General form of difference table	2 1
9	Using ($f[x_0, x_1] = (y_1 - y_0)/(x_1 - x_0)$), and ($x_1 = x_0 + h$), ⇒ ($f[x_0, x_1] = \Delta y_0/h$).	2

Q.No.	Key Points	Marks
10	(E) is shift operator; Forward (Δ) and backward (∇) difference operators related by ($\Delta = \nabla E$). Proof using operator definitions.	2
11	$y_{n+1} = y_n + h\, f(x_n, y_n)$; parameters: initial values, step size (h), derivative function (f).	2
12	From Taylor's expansion: $T_{i,j-1} = T_{i,j} - \Delta y(\partial T/\partial y) \Rightarrow (\partial T/\partial y)_{i,j} \approx (T_{i,j} - T_{i,j-1})/\Delta y$. Taylor Series expansion only	2 1
13	Employ weighted average of slopes from multiple points within each step to improve accuracy; no higher derivatives needed. Order represents the number of slopes (any two points)	2
14	Correct order: 1 → 2 → 6 → 3 → 5 → 7 → 4.	2

PART B – Short Essay Questions

Answer any 6 questions. Each carries 5 marks. [6 × 5 = 30 marks]

Q.No.	Expected Answer / Steps	Scheme of Valuation	Marks
15	Rewrite: x = (5 - y)/4; y = (6 - x)/5; Start with (0,0): Iter1: x_1=1.25, y_1=0.95; Iter2: x_2=1.0125, y_2=0.9975; Iter3: x_3=1.0006, y_3=0.9999; → Solution ≈ (1.00, 1.00).	Iterative formula (2); 3 iterations (3).	5
16	Root near ln(5)=1.609. Iterations between x=1 and x=2: f(1)=−2.282, f(2)=2.389 → mid=1.5 (f(1.5)=-0.518); f(1.75)=0.755; f(1.625)=0.078. Thus 1.625 is approximate root	Initial guess- (1) Formula-1 iterations-3	5
17	Take log: ln y = ln a + bx ⇒ linear form; normal equations for ln y vs x; compute a, b from Σx, Σln y terms. (No need to write the normal equations)	Concept (2); transformation & equations (3).	5
18	Using forward difference: Δy_0 = 1.16−1=0.16, $\Delta^2 y_0$=2.24, h=0.2; $(f'(0) \approx (1/h)(\Delta y_0 - (1/2)\Delta^2 y_0 + \ldots)) \rightarrow (1/0.2)(0.16 - 1.12) = -4.8$ (approx). Formation of the forward difference table- 2 marks	Use correct formula (3); computation (2).	5
19	(h=1); $Q = (h/3)[I_0+I_4+4(I_1+I_3)+2I_2] = (1/3)[2+4+4(2.5+3.8)+2(3.2)]$ = (1/3)(6+25.2+6.4)=12.53 C (approx).	Formula (2); substitutio	5

Q.No.	Expected Answer / Steps	Scheme of Valuation	Marks
		n (2); result with unit (1).	
20	h=2; x:0,2,4,6. y:1.000,0.200,0.0588,0.027. Integral=$(3h/8)[y_0+3(y_1+y_2)+y_3]=(3\times2/8)[1+3(0.2588)+0.027]=0.75(1.8034)=1.3526$. {True value=$\tan^{-1}6=1.405$ } If h=1; x:0,1,2,3,4,5,6; y:1.000, 0.500, 0.200,0.100, 0.0588, 0.03846, 0.027; integral=1.3570 (approx.) Any suitable choice can be considered.	Proper rule (2); computation (3) [finding y values-1.5; substituition-1, result-0.5]	5
21	$T_i^{n+1} = T_i^n + r(T_{i+1}^n - 2T_i^n + T_{i-1}^n)$; given $\Delta t=0.002$, $\Delta x=0.5 \rightarrow r=\Delta t/(\Delta x)^2=0.002/0.25=0.008$. At t=0, $T=\sin(\pi x)$: x=0,0.5,1→T=0,1,0 → $T(0.5,0.002)=1+0.008(0-2\times1+0)=0.984$.	Formulae (2); r computation (1); x and T computation-(1), substitution & result (1).	5
22	1. **Predictor (Euler Formula):** $$y_{m+1}^{(o)} = y_m + hf(x_m, y_m)$$ *(Using standard Euler method to estimate the next y value)* 2. **Corrector (Modified Euler Formula):** $$y_{m+1}^{p+1} = y_m + \frac{h}{2}\left[f(x_m, y_m) + f\left(x_{m+1}, y_{m+1}^{p}\right)\right]$$ y(0)=1, h=0.1 Predictor: $y_p=1+0.1(0+1^2)=1.1$; Corrector: $y_c=1+0.05[(0+1^2)+(0.1+1.1^2)]=1+0.05(1+1.31)=1.1155\approx1.116$. Repeated correction not required	Predictor formula-(1), Corrector formula (1) predicted value (1.5); corrector substitution-(1); final answer (0.5)	5